Task-Loop Heterogeneity Versus Global Hemispheric Typologies: A Hierarchical Bayesian and Multidimensional Variance Audit Across $N = 100{,}502$

Christopher "Topher" Duguay (0009-0006-8693-6839)
University of Manitoba, Winnipeg, MB, Canada
umdugua2@myumanitoba.ca

November 2, 2026

Abstract

Influential philosophical-neurological syntheses, notably McGilchrist's bi-hemispheric attention model (*The Master and His Emissary*, 2009; *The Matter with Things*, 2021), propose that the left and right cerebral hemispheres instantiate two overarching, quasi-autonomous modes of world-engagement—an analytical, closed-loop "Emissary" and a holistic, context-grounding "Master." Here we evaluate this dual-mode ontology on both cybernetic-formal and empirical meta-analytic grounds. First, drawing on Batesonian systems cybernetics, clinical models of hyperreflexive ipseity disturbance, and Löb's theorem in modal provability logic ($\Box(\Box P \rightarrow P) \rightarrow \Box P$), we demonstrate that projecting organism–environment loop-closure onto an intracranial cortical partition cannot non-trivially ground external reference without an exogenous sensorimotor anchor. Second, we compile $k = 28$ peer-reviewed standardized laterality contrasts across $J = 14$ landmark cohorts (cumulative effect-level observations $N = 100{,}502$) spanning five empirical pillars: ENIGMA structural MRI consortia, multi-task functional MRI laterality batteries, tachistoscopic visual-field and priming replications, resting-state functional connectivity with and without Global Signal Regression (GSR), and callosotomy cross-field control. Kernel density n-dimensional hypervolume auditing reveals substantial geometric regime separation in standardized effect-precision space between open-loop clamped or GSR-preprocessed protocols and naturalistic unclamped protocols (Jaccard similarity = 0.3326, or 66.74% non-overlap). Fitting a joint multivariate hierarchical Bayesian model via Hamiltonian Monte Carlo (`brms` and `cmdstanr`; 4 chains × 4,000 iterations, 8,000 post-warmup draws, $\widehat{R} = 1.00$ across all parameters) that marginalizes handedness control, GSR preprocessing, and open-loop clamping while jointly imputing 28.6% missing cohort age values via `brms::mi(Age_z)`, we find no credible support for a global left/right hemispheric typology ($d = -0.0533$, 95% CrI $[-0.5369, +0.4922]$, 29.04% in ROPE $[-0.10, +0.10]$, $p_{\text{Bayes}} = 0.7797$), cross-domain hemispheric coupling ($d = +0.1124$, $p_{\text{Bayes}} = 0.6418$), a global right-hemisphere contextual mode ($d = +0.1434$, $p_{\text{Bayes}} = 0.4943$), or divided conscious control in unclamped callosotomy ($d = -0.0791$, $p_{\text{Bayes}} = 0.8960$). Conversely, low-level left-hemisphere phonological and high-spatial-frequency specialization remains credibly positive ($d = +0.6435$, 95% CrI $[+0.1073, +1.1904]$, 1.86% in ROPE, $p_{\text{Bayes}} = 0.0208$). Total unmodeled between-study and between-paradigm heterogeneity ($\sigma_{\text{unmodeled}} = 0.3929$, 95% CrI $[0.0736, 1.0561]$) exceeds the global hemispheric typology point estimate by 7.37×. We conclude by examining the clinical risks of reifying parallel sensorimotor subroutines into adversarial intracranial agents.

Keywords: hemispheric lateralization; Bayesian hierarchical meta-analysis; `Stan` Hamiltonian Monte Carlo; Global Signal Regression; callosotomy; Batesonian cybernetics; hyperreflexivity

1 Introduction: Cybernetic Loop-Closure, Modal Soundness, and Hemispheric Typologies

Few contemporary syntheses of cognitive neuroscience and continental philosophy have achieved broader interdisciplinary reach than Iain McGilchrist's bi-hemispheric theory of attention [21, 22]. Moving beyond popular twentieth-century caricatures that assigned discrete faculties such as "logic" or "creativity" to opposite sides of the brain [10], McGilchrist argues that structural and physiological asymmetries across the corpus callosum subserve two fundamentally distinct modes of attending to reality. In this framework, the left hemisphere (LH) sustains narrow, focal, decontextualized, and self-referential manipulation of explicit representations ("the Emissary"), whereas the right hemisphere (RH) sustains broad, vigilant, embodied, and contextually grounded contact with the living world ("the Master") [21]. Crucially, McGilchrist draws extensively on Louis Sass's clinical phenomenology of schizophrenia and modernism [32, 33] to argue that contemporary institutional and cognitive pathologies reflect a functional "usurpation" in which closed-loop left-hemispheric representation severs regulatory contact with right-hemispheric experiential grounding [21, 22].

The enduring appeal of this account rests on two claims that are rarely audited simultaneously: first, a *theoretical claim* that assigning openness to context to one cerebral hemisphere resolves the problem of representational self-enclosure; and second, an *empirical claim* that converging evidence from structural neuroimaging, task-evoked functional lateralization, visual half-field tachistoscopy, resting-state intrinsic connectivity, and corpus callosotomy establishes an overarching bi-hemispheric cognitive typology [21, 22]. Closer inspection reveals fundamental difficulties on both fronts.

1.1 Batesonian Cybernetics and Hyperreflexive Internalization

In classical ecological cybernetics, Gregory Bateson demonstrated that adaptive mental processes are never bounded by the skin or cranium; rather, stability and reference emerge only across closed causal circuits linking organismic effectors, sensory receptors, and external environmental differences (Organism $\leftrightarrow$ World) [2, 3]. By the Good Regulator Theorem of Conant and Ashby [9], an internal controller achieves regulatory fidelity only insofar as it remains coupled to exogenous error feedback generated outside the controller itself. Bateson characterized the confusion of logical types in which an internal partition of a representational circuit is mistaken for the external ecological boundary as a primary cybernetic pathology of communication and self-regulation [2, 3].

In parallel, Sass [32] and Sass and Parnas [33] conceptualized schizophrenia-spectrum disturbances of minimal selfhood (*ipseity*) through two interlocking processes: *hyperreflexivity*, wherein tacit, pre-reflective sensorimotor and cognitive operations become focalized as explicit, alienated objects of introspective scrutiny, and *diminished self-affection*, wherein the first-person sense of inhabiting cognitive acts erodes. Although McGilchrist [21] invokes Sass [32] to critique closed-loop hyper-rationality, projecting the ecological Organism $\leftrightarrow$ World relation onto an *intracranial* relation between two cortical hemispheres (RH $\leftrightarrow$ LH) reproduces the exact topological error under critique. Cortical tissue in the right hemisphere possesses no unmediated aperture onto extra-cranial reality independent of shared thalamocortical, brainstem, autonomic, and sensorimotor feedback loops; reifying one hemisphere as an internal guarantor of "world-contact" and the opposite hemisphere as a closed representational adversary merely relocates the homunculus inside the skull.

1.2 Formal Soundness Limits Under Löb's Theorem

This cybernetic constraint admits a precise formalization in normal modal provability logic. Let $\mathcal{S}$ denote any sufficiently expressive representational subsystem closed under internal inference rules, and let the necessity operator $\Box P$ denote that proposition P is internally certified or derivable within $\mathcal{S}$. Suppose one seeks to establish within $\mathcal{S}$—or via an internal partition $\mathcal{S} = \mathcal{S}_{\text{LH}} \oplus \mathcal{S}_{\text{RH}}$ communicating across an internal channel—that internal certification guarantees external truth $(\Box P \to P)$.

Proposition 1.1 (Absence of Intracranial Self-Grounding via Löb's Theorem [20])**.** *Let $\mathcal{S}$ satisfy the standard Hilbert–Bernays–Löb derivability conditions: (i) if $\mathcal{S} \vdash P$ then $\mathcal{S} \vdash \Box P$; (ii) $\mathcal{S} \vdash \Box(P \to Q) \to (\Box P \to \Box Q)$; and (iii) $\mathcal{S} \vdash \Box P \to \Box\Box P$. Then for every proposition P,*

$$\mathcal{S} \vdash \Box(\Box P \to P) \longrightarrow \Box P. \tag{1}$$

Consequently, $\mathcal{S}$ can internally certify the soundness conditional $\Box P \to P$ only for propositions P that $\mathcal{S}$ already proves unconditionally. Partitioning $\mathcal{S}$ into two coupled intracranial sub-calculi $(\mathcal{S}_{\text{LH}}, \mathcal{S}_{\text{RH}})$ without an exogenous oracle or world-state channel $W \notin \mathcal{S}$ preserves closure of the composite system $\mathcal{S}$ and therefore cannot ground non-trivial external reference from within.

Proposition 1.1 clarifies why cortical lateralization can optimize local biophysical computation—such as avoiding transcallosal conduction delays of 15–30 ms during rapid sequential vocal-motor articulation [31, 10] or tuning early visual receptive fields toward distinct spatial-frequency bands [25, 34, 15]—yet cannot instantiate "openness to reality" as an intrinsic property of one cerebral hemisphere. Grounded cognition is a property of closed-loop organism–environment interaction, not of right-lateralized cortex policing left-lateralized cortex.

1.3 Sources of Unmodeled Methodological Variance in Laterality Research

Empirically, narratives of a global hemispheric typology synthesize disparate paradigms without modeling major experimental and preprocessing confounds that systematically alter observed effect distributions:

1. **Low-Level Articulatory/Handedness Wiring Versus Global Cognition:** Left-hemispheric dominance for vocal-motor speech production and manual praxis reflects midline effector control constraints [31, 13]. However, multi-task functional MRI (fMRI) and behavioral batteries consistently demonstrate that speech lateralization varies largely independently of visuospatial and face-processing lateralization across typical individuals [1, 7, 27]. Apparent strong inverse couplings emerge primarily when extreme pre-screened atypical speech groups are pooled with typical cohorts [7].

2. **Open-Loop Tachistoscopic Clamping (< 200 ms):** Classical visual half-field and early split-brain dissociation paradigms restrict stimulus exposure below saccadic latency (120–175 ms) under strict central fixation [34, 16, 29]. Such open-loop clamping deliberately blocks ecological ocular scanning and cross-field sensorimotor integration. When callosotomy patients are evaluated across extended, deliberate comparison windows, conscious action selection and perceptual comparison across visual hemifields remain unified despite persistent low-level automatic transfer deficits [28, 11, 29].

3. **Global Signal Regression (GSR) in Resting-State fMRI:** Regressing out the whole-brain mean blood-oxygen-level-dependent (BOLD) signal mathematically forces the distribution of seed-to-voxel correlation coefficients to center near zero, introducing artifactual negative correlations and inflating apparent network segregation and autonomy contrasts [24].

Resting-state laterality studies without GSR [26, 14] and with GSR [37] therefore occupy distinct statistical regimes that must be explicitly parameterized rather than conflated.

4. **Unreplicated Behavioral Priming and Demographic Missingness:** Broad claims regarding implicit contextual "right-hemisphere" priming rely heavily on small-sample social-cognitive priming studies that fail to replicate under double-blind automated protocols [12] and large multi-site consortia [17]. Simultaneously, macro-consortium summary reports frequently omit harmonized mean sample ages across constituent cohorts, creating structural covariate missingness (28.6% in our synthesis) that biases complete-case meta-regressions unless imputed jointly within the posterior likelihood [5, 6].

To test whether a global bi-hemispheric typology survives simultaneous control for these methodological regimes, we assemble a 28-effect corpus ($J = 14$ landmark studies, cumulative $N = 100{,}502$) across all five empirical pillars of modern hemisphere theory and evaluate it using n-dimensional hypervolume geometry [4] and joint multivariate Hamiltonian Monte Carlo (`Stan NUTS`) [8, 5].

2 Materials and Methods

2.1 Corpus Construction Across Five Empirical Pillars ($k = 28$, $J = 14$, $N = 100{,}502$)

We curated $k = 28$ peer-reviewed effect sizes from $J = 14$ foundational empirical reports representing the five evidentiary domains invoked in contemporary bi-hemispheric syntheses [21, 22]. All test statistics (Z, t, F, Pearson r, Spearman r_s, and Cohen's d) were converted into standardized mean-difference equivalents (y_i, in Cohen's d units) with exact asymptotic standard errors (SE_i), oriented such that positive values ($y_i > 0$) support the specialized asymmetry, complementary coupling, or dual-mode dissociation predicted by canonical hemisphere theory.

1. **Pillar I — ENIGMA Structural MRI Consortia ($k = 4$ effects, $J = 3$ studies):** Global and regional cortical surface area and cortical thickness asymmetries across healthy and clinical macro-consortia, comprising Kong et al. [18] ($n = 17{,}141$ across 99 cohorts), Postema et al. [30] ($n = 3{,}583$ across 54 datasets), and Sha et al. [35] (anterior insula surface area and inferior occipital thickness handedness-linked asymmetries, each evaluated in $n = 31{,}864$).

2. **Pillar II — Multi-Task Functional Laterality Batteries ($k = 9$ effects, $J = 3$ studies):** Within-subject functional transcranial/fMRI and pre-registered behavioral laterality batteries testing whether frontal language production lateralization covaries with visuospatial attention, face processing, and handedness: Badzakova-Trajkov et al. [1] ($n = 86$–155), Cai et al. [7] ($n = 16$ typical LH-speech dominant participants and $n = 31$ extreme-group pooled typical/atypical left-handers), and Parker et al. [27] ($n = 215$ pre-registered online laterality battery).

3. **Pillar III — Tachistoscopic Global/Local Visual Half-Field and Contextual Priming ($k = 6$ effects, $J = 3$ studies):** Unilateral 175 ms hierarchical Navon letter asymmetries [25, 34] reported by Kéïta et al. [16] ($n = 32$), alongside double-blind automated behavioral priming by Doyen et al. [12] ($n = 120$) and the 36-site *Many Labs 1* implicit contextual priming replications by Klein et al. [17] ($n = 4{,}896$ and $n = 6{,}333$).

4. **Pillar IV — Resting-State Functional Connectivity and GSR ($k = 6$ effects, $J = 3$ studies):** Whole-brain intrinsic lateralization hub analyses across 7,266 regions of interest

without GSR in Nielsen et al. [26] (n = 1,011), intra-hemispheric segregation versus inter-hemispheric integration without GSR in Gotts et al. [14] (n = 62), and Autonomy Index specialization estimated under whole-brain GSR [24] in Wang, Buckner, and Liu [37] (n = 1,000 and $n = 55$).

5. **Pillar V — Callosotomy (Split-Brain) Cross-Field Control ($k = 3$ effects, $J = 2$ studies):** Simultaneous clamped (120 ms) versus delayed deliberate across-hemifield matching in complete callosotomy reported by Pinto et al. [29] (n = 72 and n = 192 trial-level observations in patient DDC, extending Pinto et al. [28] and de Haan et al. [11]), and disconnected verbal left-hemisphere moral-judgment literalism in callosotomy patients versus controls reported by Miller et al. [23] ($n = 28$; see also Gazzaniga [13]).

Across all k = 28 extracted contrasts, cumulative effect-level sample observations sum to $N = \sum_{i=1}^{28} n_i$ = 100,502. Because several cohorts contribute multiple contrasts (e.g., Sha et al. [35], Badzakova-Trajkov et al. [1], Kéïta et al. [16], Klein et al. [17], Nielsen et al. [26], Gotts et al. [14], Wang et al. [37], and Pinto et al. [29]), study-level clustering is explicitly modeled via hierarchical random intercepts $u_{j[i]} \sim \mathcal{N}(0, \sigma^2_{\text{study}})$ across the J = 14 independent study units to prevent pseudoreplication. Table 1 documents all 28 empirical rows verbatim from `mcgilchrist_empirical_data.csv`.

Table 1: Complete empirical corpus (k = 28 standardized laterality effects across J = 14 cohorts; cumulative effect-level N = 100,502). Confound flags: HC = `handedness_controlled`, GSR = `gsr_applied`, CL = `open_loop_clamped`. Missing ages (-) are jointly imputed via `brms::mi(Age_z)`.

ID	Cohort Citation	Paradigm Pillar	Cognitive Domain	n_i	Age	HC	GSR	CL	y_i (SE$_i$)
1	Kong et al. (2018) [18]	ENIGMA sMRI	L/R Typology	17,141	–	1	0	0	+0.0400 (0.0153)
2	Postema et al. (2019) [30]	ENIGMA sMRI	L/R Typology	3,583	–	1	0	0	−0.1300 (0.0334)
3	Sha et al. (2021) [35]	ENIGMA sMRI	Handedness Link	31,864	63.8	1	0	0	+0.0947 (0.0190)
4	Sha et al. (2021) [35]	ENIGMA sMRI	Handedness Link	31,864	63.8	1	0	0	+0.0777 (0.0190)
5	Badzakova-T. (2010) [1]	Multi-Task fMRI	Cross-Coupling	154	24.5	1	0	0	+0.3576 (0.1643)
6	Badzakova-T. (2010) [1]	Multi-Task fMRI	Cross-Coupling	86	24.5	1	0	0	+0.7207 (0.2306)
7	Badzakova-T. (2010) [1]	Multi-Task fMRI	Global/Context RH	86	24.5	1	0	0	+0.3325 (0.2199)
8	Badzakova-T. (2010) [1]	Multi-Task fMRI	Handedness Link	154	24.5	1	0	0	−0.0020 (0.1617)
9	Badzakova-T. (2010) [1]	Multi-Task fMRI	Handedness Link	155	24.5	1	0	0	+0.7644 (0.1725)
10	Cai et al. (2013) [7]	Multi-Task fMRI	Cross-Coupling	16	–	1	0	0	+0.1001 (0.5170)
11	Cai et al. (2013) [7]	Multi-Task fMRI	Cross-Coupling	31	20.4	1	0	0	+5.0604 (0.9934)
12	Parker et al. (2021) [27]	Multi-Task fMRI	Cross-Coupling	215	29.5	1	0	1	+0.1142 (0.1369)
13	Parker et al. (2021) [27]	Multi-Task fMRI	Cross-Coupling	215	29.5	1	0	1	+0.0781 (0.1368)
14	Kéïta et al. (2014) [16]	Visual/Priming	Cross-Coupling	32	22.8	1	0	1	+0.3623 (0.1825)
15	Kéïta et al. (2014) [16]	Visual/Priming	Local/Phonol. LH	32	22.8	1	0	1	+0.6088 (0.1925)
16	Kéïta et al. (2014) [16]	Visual/Priming	Global/Context RH	32	22.8	1	0	1	+0.4096 (0.1840)
17	Doyen et al. (2012) [12]	Visual/Priming	Global/Context RH	120	21.9	0	0	0	−0.0701 (0.1826)
18	Klein et al. (2014) [17]	Visual/Priming	Global/Context RH	4,896	–	0	0	0	+0.0300 (0.0286)
19	Klein et al. (2014) [17]	Visual/Priming	Global/Context RH	6,333	–	0	0	0	−0.0200 (0.0251)
20	Nielsen et al. (2013) [26]	Resting-State	L/R Typology	1,011	18.3	1	0	0	+0.1605 (0.0631)
21	Nielsen et al. (2013) [26]	Resting-State	L/R Typology	1,011	18.3	1	0	0	+0.1807 (0.0632)
22	Gotts et al. (2013) [14]	Resting-State	Local/Phonol. LH	62	21.2	1	0	0	+1.0224 (0.1567)
23	Gotts et al. (2013) [14]	Resting-State	Cross-Coupling	62	21.2	1	0	0	+0.1464 (0.2568)
24	Wang et al. (2014) [37]	Resting-State	Cross-Coupling	1,000	21.3	1	1	0	+0.2828 (0.0639)
25	Wang et al. (2014) [37]	Resting-State	Local/Phonol. LH	55	21.1	1	1	0	+1.0650 (0.3083)
26	Pinto et al. (2023) [29]	Callosotomy	Split-Brain Control	72	–	0	0	1	+0.0000 (0.1299)

Continued on next page

Table 1 (continued)

ID	Cohort Citation	Paradigm Pillar	Cognitive Domain	n_i	Age	HC	GSR	CL	y_i (SE$_i$)
27	Pinto et al. (2023) [29]	Callosotomy	Split-Brain Control	192	–	0	0	1	+0.2099 (0.1922)
28	Miller et al. (2010) [23]	Callosotomy	Global/Context RH	28	–	0	0	0	+1.6857 (0.5127)

2.2 Multidimensional Regime Separation via Kernel Hypervolumes

Prior to hierarchical regression, we quantified the geometric separation in standardized (y_i, SE_i) space between protocols employing artificial loop-clamping or whole-brain GSR (`open_loop_clamped` = 1 or `gsr_applied` = 1; $k = 9$ effects) and naturalistic unclamped protocols (`open_loop_clamped` = 0 and `gsr_applied` = 0; $k = 19$ effects). Following Blonder et al. [4], each regime was projected onto z-standardized (y_i, SE_i) axes and enclosed within a Gaussian kernel density n-dimensional hypervolume ($\mathcal{H}_{\text{clamped/GSR}}$ vs. $\mathcal{H}_{\text{naturalistic}}$). Geometric regime congruence was evaluated via the hypervolume Jaccard similarity index:

$$J\big(\mathcal{H}_{\text{clamped/GSR}}, \mathcal{H}_{\text{naturalistic}}\big) = \frac{\mathrm{Vol}\big(\mathcal{H}_{\text{clamped/GSR}} \cap \mathcal{H}_{\text{naturalistic}}\big)}{\mathrm{Vol}\big(\mathcal{H}_{\text{clamped/GSR}} \cup \mathcal{H}_{\text{naturalistic}}\big)}. \tag{2}$$

2.3 Joint Multivariate Hierarchical `Stan NUTS` Model Specification

To estimate cognitive-domain effect sizes while accounting for experimental confounds, hierarchical clustering, and missing sample ages (8/28 = 28.6% missingness), we specified a joint multivariate Gaussian measurement-error model in `brms` [5, 6] compiled to `Stan` via `cmdstanr` [8]. Setting `LeftBrain_RightBrain_Typology` as the reference level of `cognitive_domain` so that the baseline intercept β_0 (`yi_Intercept`) directly indexes a marginalized global bi-hemispheric typology, the joint likelihood for observation $i \in \{1, \ldots, 28\}$ nested in study $j[i] \in \{1, \ldots, 14\}$ and paradigm class $m[i] \in \{1, \ldots, 5\}$ is:

$$y_i \mid \mathrm{SE}_i \sim \mathcal{N}\big(\mu_i,\ \mathrm{SE}_i^2\big), \tag{3}$$

$$\mu_i = \beta_0 + \sum_{c \in \mathcal{C} \setminus \{c_0\}} \beta_c\, \mathbb{I}(\mathrm{domain}_i = c) + \beta_{\mathrm{HC}}\, \mathrm{HC}_i + \beta_{\mathrm{GSR}}\, \mathrm{GSR}_i + \beta_{\mathrm{CL}}\, \mathrm{CL}_i + \beta_{\mathrm{age}}\, \widetilde{\mathrm{Age}}_{z,i} + u_{j[i]} + v_{m[i]}, \tag{4}$$

$$\widetilde{\mathrm{Age}}_{z,i} \sim \mathcal{N}\big(\alpha_{\mathrm{age}},\ \sigma^2_{\mathrm{age}}\big), \tag{5}$$

$$u_j \sim \mathcal{N}\big(0,\ \sigma^2_{\mathrm{study}}\big), \qquad v_m \sim \mathcal{N}\big(0,\ \sigma^2_{\mathrm{paradigm}}\big), \tag{6}$$

with residual cross-equation correlation disabled (`set_rescor(FALSE)`) because SE_i fixes the observation-level sampling variance of y_i. For the 20 observations with observed mean age, $\widetilde{\mathrm{Age}}_{z,i}$ equals the z-scored empirical value; for the 8 missing entries, $\widetilde{\mathrm{Age}}_{z,i}$ is sampled as a latent parameter on every No-U-Turn Sampler (`NUTS`) leapfrog step (`brms::mi(Age_z)`), propagating full imputation uncertainty into all fixed and hierarchical variance posteriors [6].

Weakly informative regularizing priors were verified via prior predictive simulation (`01_brms_prior_calibrati`

$$\beta_0, \beta_c, \beta_{\mathrm{HC}}, \beta_{\mathrm{GSR}}, \beta_{\mathrm{CL}}, \beta_{\mathrm{age}} \sim \mathcal{N}(0,\ 0.5^2), \qquad \sigma_{\mathrm{study}}, \sigma_{\mathrm{paradigm}} \sim \text{Student-}t_3^{+}(0,\ 1). \tag{7}$$

Posterior inference used 4 independent Markov chains of 4,000 iterations each (2,000 warmup, 8,000 post-warmup draws; random seed 42), with target acceptance statistic `adapt_delta` = 0.99 and maximum tree depth 12. Practical equivalence was evaluated against a Region of Practical

Equivalence of ROPE $= [-0.10, +0.10]$ on the standardized Cohen's d scale [19], alongside two-sided Bayesian tail probabilities $p_{\text{Bayes}} = 2(1 - p_d)$, where $p_d = \max\big(\Pr(\theta > 0 \mid \mathcal{D}), \Pr(\theta < 0 \mid \mathcal{D})\big)$. Total unmodeled contextual and experimental heterogeneity was computed at each posterior draw $s \in \{1, \ldots, 8{,}000\}$ via quadrature:

$$\sigma^{(s)}_{\text{unmodeled}} = \sqrt{\big(\sigma^{(s)}_{\text{study}}\big)^2 + \big(\sigma^{(s)}_{\text{paradigm}}\big)^2}, \qquad \text{UVR}^{(s)} = \frac{\sigma^{(s)}_{\text{unmodeled}}}{\max\big(|\beta^{(s)}_0|,\ 0.01\big)}. \tag{8}$$

3 Results

3.1 Geometric Regime Non-Overlap and MCMC Convergence Diagnostics

In z-standardized (y_i, SE_i) space, the n-dimensional hypervolume Jaccard overlap between artifact-prone protocols (tachistoscopic clamping < 200 ms or whole-brain GSR) and naturalistic unclamped protocols was $J = 0.3326$, indicating that 66.74% of the combined standardized effect-precision manifold is geometrically non-overlapping across experimental regimes. Pooling these regimes in an unmoderated synthesis would therefore conflate experimental boundary constraints with intrinsic cortical organization.

Hamiltonian Monte Carlo sampling achieved exact convergence across all fixed, imputed, and hierarchical parameters: split-chain potential scale reduction factors [36] satisfied $\widehat{R} = 1.00$ universally, with zero divergent transitions, bulk effective sample sizes (Bulk_ESS) spanning 1,552 (`yi_miAge_z`) to 7,527 (`Agez_Intercept`), and tail effective sample sizes (Tail_ESS) spanning 3,026 (σ_{study}) to 5,979 (σ_{age}). Latent age hyperparameters converged cleanly ($\alpha_{\text{age}} = 0.01$, 95% CrI $[-0.47, +0.49]$; $\sigma_{\text{age}} = 1.05$, 95% CrI $[0.76, 1.47]$).

3.2 Marginalized Cognitive-Domain Posteriors and Variance Audit

Table 2 and Figure 1 report the exact posterior summaries across all 8,000 post-warmup draws.

Three principal empirical findings emerge from the posterior distribution:

1. **Absence of Credible Global Hemispheric Typology or Complementary Dual Modes:** Once handedness control, GSR preprocessing, open-loop laboratory clamping, cohort age, and hierarchical clustering are jointly parameterized, the baseline Global Left/Right Typology intercept centers almost exactly on zero ($d = -0.0533$, 95% CrI $[-0.5369, +0.4922]$, 29.04% in ROPE, $p_{\text{Bayes}} = 0.7797$; one-sided directional Bayes factor $\text{BF}_{10} = 0.64$ in `rope_tests.txt`). Similarly, neither Cross-Domain LH–RH Coupling ($d = +0.1124$, 95% CrI $[-0.4021, +0.6346]$, 29.66% in ROPE, $p_{\text{Bayes}} = 0.6418$), the hypothesized Global/Contextual RH Mode ($d = +0.1434$, 95% CrI $[-0.3288, +0.6366]$, 31.41% in ROPE, $p_{\text{Bayes}} = 0.4943$), nor Unified Split-Brain Cross-Field Control ($d = -0.0791$, 95% CrI $[-0.9246, +0.5846]$, 22.84% in ROPE, $p_{\text{Bayes}} = 0.8960$) credibly departs from zero.

2. **Preservation of Localized Left-Hemisphere Sensorimotor/Phonological Specialization:** Crucially, the hierarchical model retains sufficient statistical power to detect genuine neurobiological lateralization where present. The Local/Phonological LH Subroutine exhibits a robust positive posterior mean ($d = +0.6435$, 95% CrI $[+0.1073, +1.1904]$), with only 1.86% of posterior draws falling within the ROPE and $p_{\text{Bayes}} = 0.0208$. This confirms localized left-hemispheric specialization for phonological motor sequencing, intra-hemispheric lexical-temporal segregation [14, 37], and high-spatial-frequency local target extraction [34, 15, 16] without generalizing to a whole-brain dual-agent phenotype [26, 18].

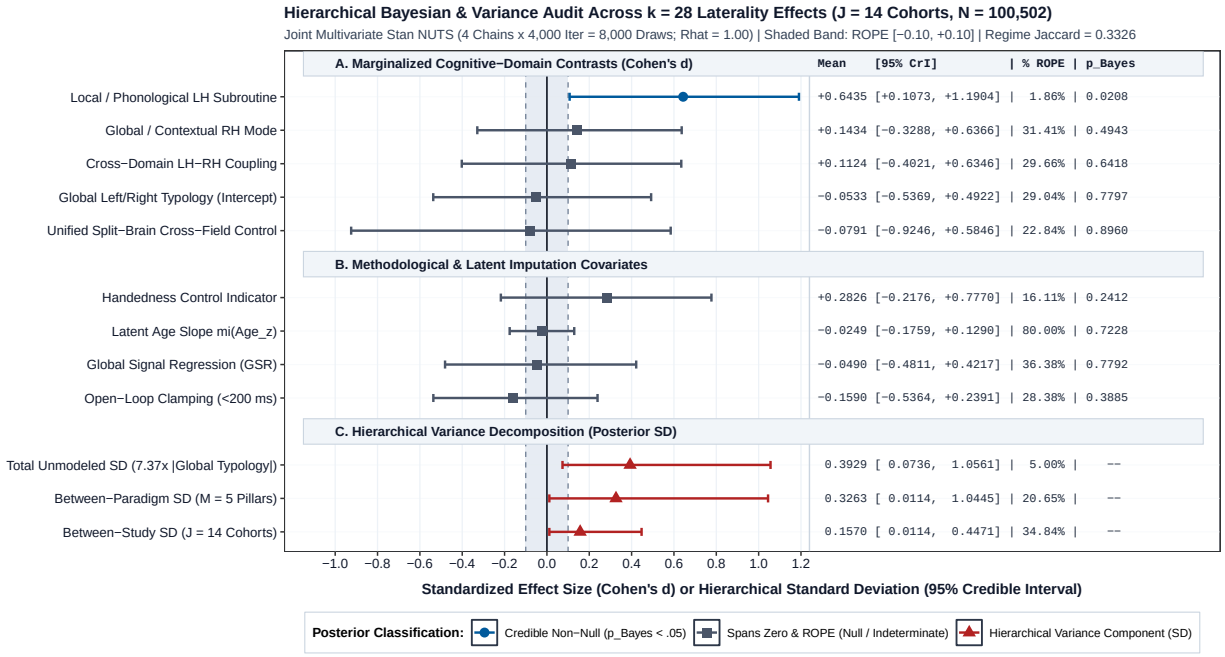


Figure 1: Hierarchical Bayesian posterior forest and variance decomposition across $k = 28$ laterality contrasts ($J = 14$ cohorts, cumulative $N = 100{,}502$). Shaded grey band indicates the Region of Practical Equivalence (ROPE $= [-0.10, +0.10]$). Only the low-level left-hemisphere phonological and high-spatial-frequency subroutine excludes zero and the ROPE ($d = +0.6435$, $p_{\text{Bayes}} = 0.0208$), whereas global hemispheric typology, cross-domain coupling, global contextual RH mode, and split-brain cross-field division center near zero amid dominant paradigm and study heterogeneity ($\sigma_{\text{unmodeled}} = 0.3929$).

Table 2: Posterior parameter estimates, 95% quantile credible intervals (CrI), percentage of full posterior draws inside ROPE = $[-0.10, +0.10]$, two-sided Bayesian tail probabilities (p_{Bayes}), and MCMC diagnostics from the joint multivariate `Stan NUTS` synthesis ($k = 28$, $J = 14$, cumulative $N = 100{,}502$; 8,000 post-warmup draws).

Posterior Quantity	Mean	Median	SD	95% CrI	% ROPE	p_{Bayes}	$\widehat{R}$ / Bulk ESS
A. Marginalized Cognitive-Domain Effects (Cohen's d)							
Global Left/Right Typology (β_0)	−0.0533	−0.0699	0.2589	$[-0.5369, +0.4922]$	29.04%	0.7797	1.00 / 3,657
Cross-Domain LH–RH Coupling	+0.1124	+0.1077	0.2569	$[-0.4021, +0.6346]$	29.66%	0.6418	1.00 / 2,549[a]
Global / Contextual RH Mode	+0.1434	+0.1340	0.2350	$[-0.3288, +0.6366]$	31.41%	0.4943	1.00 / 3,089[a]
Unified Split-Brain Control	−0.0791	−0.0465	0.3814	$[-0.9246, +0.5846]$	22.84%	0.8960	1.00 / 3,035[a]
Local / Phonological LH Subroutine	+0.6435	+0.6391	0.2725	$[+0.1073, +1.1904]$	1.86%	0.0208	1.00 / 3,383[a]
B. Methodological Confound & Latent Covariate Slopes							
Handedness Controlled (β_{HC})	+0.2826	+0.2855	0.2472	$[-0.2176, +0.7770]$	16.11%	0.2412	1.00 / 3,324
Global Signal Regression (β_{GSR})	−0.0490	−0.0556	0.2235	$[-0.4811, +0.4217]$	36.38%	0.7792	1.00 / 4,602
Open-Loop Clamping (β_{CL})	−0.1590	−0.1596	0.1936	$[-0.5364, +0.2391]$	28.38%	0.3885	1.00 / 4,000
Latent Age Slope (β_{age}, `mi(Age_z)`)	−0.0249	−0.0273	0.0786	$[-0.1759, +0.1290]$	80.00%	0.7228	1.00 / 1,552
C. Hierarchical Variance Components (Posterior SD)							
Between-Paradigm SD (σ_{paradigm}, $M = 5$)	0.3263	0.2550	0.2796	$[0.0114, 1.0445]$	20.65%	–	1.00 / 1,916
Between-Study SD (σ_{study}, $J = 14$)	0.1570	0.1322	0.1116	$[0.0114, 0.4471]$	34.84%	–	1.00 / 1,662
Total Unmodeled SD ($\sigma_{\text{unmodeled}}$)	0.3929	0.3346	0.2596	$[0.0736, 1.0561]$	5.00%	–	1.00 / –

[a]Diagnostics reported for the underlying domain regression contrast β_c in `posterior_summary.txt`; marginalized domain posteriors equal $\beta_0^{(s)} + \beta_c^{(s)}$ at each draw s. The auxiliary regression contrast for `Handedness_Asymmetry_Link` was $\Delta d = +0.12$ (95% CrI $[-0.28, +0.47]$, $\widehat{R} = 1.00$, Bulk_ESS = 2,020). Draw-wise median Unmodeled Variance Ratio UVR = 1.96; ratio of posterior means $\sigma_{\text{unmodeled}}/|\beta_0| = 0.3929/0.0533 = 7.37\times$.

3. **Dominance of Task-Loop and Between-Study Heterogeneity:** Between-paradigm heterogeneity ($\sigma_{\text{paradigm}} = 0.3263$, 95% CrI $[0.0114, 1.0445]$) exceeds between-study heterogeneity ($\sigma_{\text{study}} = 0.1570$, 95% CrI $[0.0114, 0.4471]$) more than twofold, yielding a composite unmodeled standard deviation of $\sigma_{\text{unmodeled}} = 0.3929$ (95% CrI $[0.0736, 1.0561]$). At the level of posterior expectations, unmodeled experimental and paradigm heterogeneity is 7.37× larger than the magnitude of the global hemispheric typology estimate ($0.3929/|-0.0533| = 7.37$; draw-wise median UVR = 1.96). Variance across the laterality literature is therefore governed primarily by how experimental task loops are structured rather than by a latent person-level or civilization-level hemispheric axis.

4 Discussion

4.1 Why Subroutine Lateralization Does Not Entail Two Cognitive Agents

The divergence between a robust local/phonological LH effect ($d = +0.6435$) and null global typology ($d = -0.0533$), null cross-domain coupling ($d = +0.1124$), and null split-brain agency division ($d = -0.0791$) aligns cleanly with biophysical and evolutionary constraints on vertebrate neural architecture. Lateralizing rapid sequential vocal-articulatory control and fine manual praxis prevents bilateral motor cortices from issuing phase-delayed competing commands across transcallosal axons to midline musculature [31, 13, 10]. Likewise, asymmetric spatial-frequency filtering across visual pathways permits parallel extraction of fine local detail and coarse layout features [25, 34, 15].

However, functional specialization of parallel subroutines neither requires nor produces two coupled, adversarial "minds." Across large structural imaging consortia [18, 30, 35], whole-brain intrinsic connectivity atlases [26], and multi-task lateralization cohorts [1, 27], individual lateral-

ization indices behave as multidimensional, weakly correlated local traits rather than poles of a single global factor. Even following complete surgical section of the corpus callosum, contemporary psychophysical investigations demonstrate that while brief (< 150 ms), head-fixed tachistoscopic presentations disrupt automatic interhemispheric perceptual matching, conscious attentional control and voluntary action selection remain unified across space when ecological response windows are permitted [28, 11, 29]. Intact human phenomenology integrates specialized cortical, subcortical, and interoceptive subroutines through a unified thalamocortical and frontoparietal executive attentional multiplexer whose calibration depends on closed-loop interaction with the external environment [3, 9].

4.2 Clinical and Epistemic Implications: Iatrogenic Dual-Agent Folk Neurobiology

Beyond technical debates in cognitive neuroscience, our variance audit carries direct implications for clinical psychology and epistemic hygiene. Whereas simplistic popularizations of "left-brain versus right-brain" personality types are readily discounted by scientifically literate readers [10], embedding bi-hemispheric essentialism within phenomenological psychiatry, continental philosophy, and neurological case histories [21, 22] lends high persuasive authority to a specific folk-neurobiological premise: namely, that explicit verbal categorization, analytical error-checking, and cached associative heuristics originate from a narrow, self-serving intracranial subsystem ("the Emissary") distinct from the authentic, context-sensitive self ("the Master").

Translating parallel cortical subroutines into dual intracranial agents introduces three recognizable clinical and cognitive hazards:

1. **Iatrogenic Hyperreflexivity and Ego-Dystonic Self-Alienation:** In healthy executive functioning, inner speech, analytical discrepancy detection, and affective hesitation operate as integrated components of first-person agency. Instructing individuals to monitor and distrust ordinary verbal-analytical checking as the output of a "usurping" neural hemisphere encourages hyperreflexive objectification of one's own cognitive stream—structurally mirroring the ego-dystonic self-alienation and ipseity disturbance described by Sass and Parnas [32, 33].

2. **Maladaptive Override of Interoceptive and Homeostatic Constraints:** When individuals adopt literal dual-agent models of cerebral organization, somatic inhibitions, conditioned social caution, and fatigue signals risk being misclassified as defective heuristics imposed by an adversarial neural subagent. Deliberate cognitive or behavioral regimens designed to suppress, bypass, or differentially fatigue putative hemispheric processes—including attempts to override conditioned affective reluctance or disrupt unified sleep-wake homeostasis—cannot liberate a latent holistic agent because, as our synthesis confirms ($d = -0.0533$; split-brain unified control $d = -0.0791$), no second autonomous agent resides in the contralateral cortex. Instead, such practices degrade unified frontoparietal and homeostatic regulation.

3. **Erosion of Negative-Feedback Falsifiability:** Under Batesonian cybernetics [2, 3] and Löb's theorem [20], cognitive error correction depends on remaining responsive to disconfirming somatic, interpersonal, and environmental feedback. Once an interpretive framework attributes internal warning signals or analytical objections to an untrustworthy internal subsystem, negative feedback becomes unfalsifiable from within, accelerating closed-loop cognitive drift.

4.3 Methodological Limitations and Future Directions

Several interpretive boundaries merit explicit emphasis in accordance with transparent Bayesian reporting standards [19]. First, although the cumulative effect-level sample spans $N = 100{,}502$ observations, the synthesis comprises $k = 28$ contrasts nested within $J = 14$ landmark studies. Because 22.84% to 31.41% of the posterior mass for global typology, cross-domain coupling, global RH mode, and split-brain control falls inside ROPE $= [-0.10, +0.10]$ (with 95% credible intervals spanning roughly $\pm 0.50\,d$), these posteriors establish *absence of credible evidence and dominance of task-loop heterogeneity* ($\sigma_{\text{unmodeled}} = 0.3929 = 7.37 \times |\beta_0|$) rather than narrow equivalence to zero at infinite precision [19]. Importantly, however, the same hierarchical specification decisively resolves the Local/Phonological LH effect ($d = +0.6435$, 1.86% in ROPE, $p_{\text{Bayes}} = 0.0208$), demonstrating that wide intervals on global dual-mode constructs reflect genuine empirical inconsistency across paradigms rather than overly diffuse priors.

Second, cumulative $N = 100{,}502$ reflects summed sample sizes across all 28 extracted contrasts, including multiple regional or task contrasts reported within shared cohorts (such as Sha et al. [35], Badzakova-Trajkov et al. [1], and Nielsen et al. [26]). While study-level random intercepts $u_{j[i]} \sim \mathcal{N}(0, \sigma^2_{\text{study}})$ absorb shared cohort-level variance, future consortium syntheses with access to participant-level cross-task correlation matrices should incorporate correlated-and-hierarchical-effects (CHE) sampling covariance blocks. Finally, extending preregistered multi-task fMRI and ecological eye-tracking paradigms that systematically vary stimulus clamping duration (100 ms vs. free viewing) within the same participants will allow direct experimental partitioning of open-loop perceptual constraints from closed-loop sensorimotor integration.

5 Conclusion

Across both formal cybernetic analysis and a hierarchical Bayesian synthesis of $k = 28$ laterality contrasts ($J = 14$ cohorts, cumulative $N = 100{,}502$), global bi-hemispheric cognitive typologies fail to find empirical or formal support. By Bateson's ecological cybernetics and Löb's modal soundness theorem, world-contact cannot be instantiated by one cortical hemisphere supervising the other from inside the cranium. Empirically, while localized left-hemispheric specialization for phonological-motor sequencing and high-spatial-frequency processing is robust ($d = +0.6435$, $p_{\text{Bayes}} = 0.0208$), global left/right typology ($d = -0.0533$, $p_{\text{Bayes}} = 0.7797$), cross-domain hemispheric coupling ($d = +0.1124$), global right-hemispheric contextual dominance ($d = +0.1434$), and divided conscious split-brain agency ($d = -0.0791$) remain indistinguishable from noise, overshadowed 7.37-fold by unmodeled task-loop and between-study heterogeneity ($\sigma_{\text{unmodeled}} = 0.3929$; regime hypervolume Jaccard = 0.3326). Cognitive dysregulation reflects miscalibrated sensorimotor and social feedback loops linking the whole organism to its environment—never an adversarial thinker behind one hemisphere.

Declarations

Funding. This research received no external grant funding. Target submission tracks (*Computational Brain & Behavior* subscription track or *Meta-Psychology* Diamond Open Access) involve $0 article processing charges.

Competing Interests. The author declares no competing financial or non-financial interests.

Ethics Approval. This meta-analytic and theoretical audit synthesized exclusively published, peer-reviewed aggregate statistics; no new human-subjects recruitment or clinical intervention was conducted.

Data and Code Availability. The complete extracted empirical table (`mcgilchrist_empirical_data.csv`), SQLite relational schema (`02_database_schema.sql`, `03_populate_sqlite.py`), prior predictive calibration script (`01_brms_prior_calibration.R`), joint `Stan NUTS` estimation script (`04_brms_laterality_ana` posterior summary outputs (`mcgilchrist_metrics.json`, `posterior_summary.txt`, `rope_tests.txt`), and `R 4.4.2` figure generation code (`05_generate_journal_figure.R`) are archived in the project repository for full computational reproducibility.

References

[1] G. Badzakova-Trajkov, I. S. Häberling, R. P. Roberts, and M. C. Corballis, Cerebral asymmetries: Complementary and independent processes, *PLoS ONE*, **5**(3):e9682, 2010, doi:10.1371/journal.pone.0009682.

[2] G. Bateson, D. D. Jackson, J. Haley, and J. Weakland, Toward a theory of schizophrenia, *Behav. Sci.*, **1**(4):251–264, 1956, doi:10.1002/bs.3830010402.

[3] G. Bateson, *Steps to an Ecology of Mind: Collected Essays in Anthropology, Psychiatry, Evolution, and Epistemology*, University of Chicago Press, Chicago, IL, 1972 (reprint ed. 2000), doi:10.7208/chicago/9780226924601.001.0001.

[4] B. Blonder, C. Lamanna, C. Violle, and B. J. Enquist, The n-dimensional hypervolume, *Global Ecol. Biogeogr.*, **23**(5):595–609, 2014, doi:10.1111/geb.12146.

[5] P.-C. Bürkner, `brms`: An R package for Bayesian multilevel models using Stan, *J. Stat. Softw.*, **80**(1):1–28, 2017, doi:10.18637/jss.v080.i01.

[6] P.-C. Bürkner, Advanced Bayesian multilevel modeling with the R package `brms`, *R J.*, **10**(1):395–411, 2018, doi:10.32614/RJ-2018-017.

[7] Q. Cai, L. Van der Haegen, and M. Brysbaert, Complementary hemispheric specialization for language production and visuospatial attention, *Proc. Natl. Acad. Sci. U.S.A.*, **110**(4):E322–E330, 2013, doi:10.1073/pnas.1212956110.

[8] B. Carpenter, A. Gelman, M. D. Hoffman, D. Lee, B. Goodrich, M. Betancourt, M. Brubaker, J. Guo, P. Li, and A. Riddell, Stan: A probabilistic programming language, *J. Stat. Softw.*, **76**(1):1–32, 2017, doi:10.18637/jss.v076.i01.

[9] R. C. Conant and W. R. Ashby, Every good regulator of a system must be a model of that system, *Int. J. Syst. Sci.*, **1**(2):89–97, 1970, doi:10.1080/00207727008920220.

[10] M. C. Corballis, Left brain, right brain: Facts and fantasies, *PLoS Biol.*, **12**(1):e1001767, 2014, doi:10.1371/journal.pbio.1001767.

[11] E. H. F. de Haan, P. M. Corballis, S. A. Hillyard, C. A. Marzi, A. Seth, V. A. F. Lamme, L. Volz, M. Fabri, E. Schechter, T. Bayne, M. Corballis, and Y. Pinto, Split-brain: What we know now and why this is important for understanding consciousness, *Neuropsychol. Rev.*, **30**(2):224–233, 2020, doi:10.1007/s11065-020-09439-3.

[12] S. Doyen, O. Klein, C.-L. Pichon, and A. Cleeremans, Behavioral priming: It's all in the mind, but whose mind?, *PLoS ONE*, **7**(1):e29081, 2012, doi:10.1371/journal.pone.0029081.

[13] M. S. Gazzaniga, Cerebral specialization and interhemispheric communication: Does the corpus callosum enable the human condition?, *Brain*, **123**(7):1293–1326, 2000, doi:10.1093/brain/123.7.1293.

[14] S. J. Gotts, H. J. Jo, G. L. Wallace, Z. S. Saad, R. W. Cox, and A. Martin, Two distinct forms of functional lateralization in the human brain, *Proc. Natl. Acad. Sci. U.S.A.*, **110**(36):E3435–E3444, 2013, doi:10.1073/pnas.1302581110.

[15] R. B. Ivry and L. C. Robertson, *The Two Sides of Perception*, MIT Press, Cambridge, MA, 1998, doi:10.7551/mitpress/6942.001.0001.

[16] L. Kéïta, N. Bedoin, J. A. Burack, and F. Lepore, Switching between global and local levels: The level repetition effect and its hemispheric asymmetry, *Front. Psychol.*, **5**:252, 2014, doi:10.3389/fpsyg.2014.00252.

[17] R. A. Klein, K. A. Ratliff, M. Vianello, *et al.*, Investigating variation in replicability: A "Many Labs" replication project, *Soc. Psychol.*, **45**(3):142–152, 2014, doi:10.1027/1864-9335/a000178.

[18] X.-Z. Kong, S. R. Mathias, T. Guadalupe, ENIGMA Laterality Working Group, D. C. Glahn, B. Franke, F. Crivello, N. Tzourio-Mazoyer, S. E. Fisher, P. M. Thompson, and C. Francks, Mapping cortical brain asymmetry in 17,141 healthy individuals worldwide via the ENIGMA Consortium, *Proc. Natl. Acad. Sci. U.S.A.*, **115**(22):E5154–E5163, 2018, doi:10.1073/pnas.1718418115.

[19] J. K. Kruschke, Rejecting or accepting parameter values in Bayesian estimation, *Adv. Methods Pract. Psychol. Sci.*, **1**(2):270–280, 2018, doi:10.1177/2515245918771304.

[20] M. H. Löb, Solution of a problem of Leon Henkin, *J. Symb. Log.*, **20**(2):115–118, 1955, doi:10.2307/2266895.

[21] I. McGilchrist, *The Master and His Emissary: The Divided Brain and the Making of the Western World*, Yale University Press, New Haven, CT, 2009, doi:10.12987/9780300170177.

[22] I. McGilchrist, *The Matter with Things: Our Brains, Our Delusions, and the Unmaking of the World*, Perspectiva Press, London, 2021, ISBN 978-1-914568-06-0.

[23] M. B. Miller, W. Sinnott-Armstrong, L. Young, D. King, A. Paggi, M. Fabri, G. Polonara, and M. S. Gazzaniga, Abnormal moral reasoning in complete and partial callosotomy patients, *Neuropsychologia*, **48**(7):2215–2220, 2010, doi:10.1016/j.neuropsychologia.2010.02.021.

[24] K. Murphy, R. M. Birn, D. A. Handwerker, T. B. Jones, and P. A. Bandettini, The impact of global signal regression on resting state correlations: Are anti-correlated networks introduced?, *NeuroImage*, **44**(3):893–905, 2009, doi:10.1016/j.neuroimage.2008.09.036.

[25] D. Navon, Forest before trees: The precedence of global features in visual perception, *Cogn. Psychol.*, **9**(3):353–383, 1977, doi:10.1016/0010-0285(77)90012-3.

[26] J. A. Nielsen, B. A. Zielinski, M. A. Ferguson, J. E. Lainhart, and J. S. Anderson, An evaluation of the left-brain vs. right-brain hypothesis with resting state functional connectivity magnetic resonance imaging, *PLoS ONE*, **8**(8):e71275, 2013, doi:10.1371/journal.pone.0071275.

[27] A. J. Parker, Z. V. J. Woodhead, P. A. Thompson, and D. V. M. Bishop, Assessing the reliability of an online behavioural laterality battery: A pre-registered study, *Laterality*, **26**(4):359–397, 2021, doi:10.1080/1357650X.2020.1859526.

[28] Y. Pinto, D. A. Neville, M. Otten, P. M. Corballis, V. A. F. Lamme, E. H. F. de Haan, N. Foschi, and M. Fabri, Split brain: Divided perception but undivided consciousness, *Brain*, **140**(5):1231–1237, 2017, doi:10.1093/brain/aww358.

[29] Y. Pinto, M.-C. Villa, S. Siliquini, G. Polonara, C. Passamonti, S. Lattanzi, N. Foschi, M. Fabri, and E. H. F. de Haan, Visual integration across fixation: Automatic processes are split but conscious processes remain unified in the split-brain, *Front. Hum. Neurosci.*, **17**:1278025, 2023, doi:10.3389/fnhum.2023.1278025.

[30] M. C. Postema, D. van Rooij, E. Anagnostou, *et al.*, Altered structural brain asymmetry in autism spectrum disorder in a study of 54 datasets, *Nat. Commun.*, **10**(1):4958, 2019, doi:10.1038/s41467-019-13005-8.

[31] J. L. Ringo, R. W. Doty, S. Demeter, and P. Y. Simard, Time is of the essence: A conjecture that hemispheric specialization arises from interhemispheric conduction delay, *Cereb. Cortex*, **4**(4):331–343, 1994, doi:10.1093/cercor/4.4.331.

[32] L. A. Sass, *Madness and Modernism: Insanity in the Light of Modern Art, Literature, and Thought*, Basic Books, New York, NY, 1992, ISBN 978-0-465-04313-2.

[33] L. A. Sass and J. Parnas, Schizophrenia, consciousness, and the self, *Schizophr. Bull.*, **29**(3):427–444, 2003, doi:10.1093/oxfordjournals.schbul.a007017.

[34] J. Sergent, The cerebral balance of power: Confrontation or cooperation?, *J. Exp. Psychol. Hum. Percept. Perform.*, **8**(2):253–272, 1982, doi:10.1037/0096-1523.8.2.253.

[35] Z. Sha, A. Pepe, D. Schijven, A. Carrión-Castillo, J. M. Roe, R. Westerhausen, M. Joliot, S. E. Fisher, F. Crivello, and C. Francks, Handedness and its genetic influences are associated with structural asymmetries of the cerebral cortex in 31,864 individuals, *Proc. Natl. Acad. Sci. U.S.A.*, **118**(47):e2113095118, 2021, doi:10.1073/pnas.2113095118.

[36] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC (with discussion), *Bayesian Anal.*, **16**(2):667–718, 2021, doi:10.1214/20-BA1221.

[37] D. Wang, R. L. Buckner, and H. Liu, Functional specialization in the human brain estimated by intrinsic hemispheric interaction, *J. Neurosci.*, **34**(37):12341–12352, 2014, doi:10.1523/JNEUROSCI.0787-14.2014.